

[DC] Neck Muscles in Head-Mounted Displays Interaction

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ABSTRACT

Head movement is the central of head-mounted displays (HMDs) interaction. Neck muscles drive head movement and play a crucial role in the body's proprioception. However, few studies have investigated how neck muscle can affect HMD interaction. This PhD research aims to explore and leverage the knowledge of neck muscles to improve HMD interaction experience. In the current progress, we have successfully leveraged the neck muscle understanding to improve head amplification. Our primary study results show that the novel head amplification can reduce neck discomfort without creating significant motion sickness. However, further research is necessary to explore how other interaction techniques and methods can be refined and improved. This PhD research introduces a novel direction to improve the HMD interaction experience. By using insight into the neck muscles, we have the potential to further improve HMD interaction experience.

Index Terms: Virtual Reality, Head-Mounted Displays, Ergonomics, Head Movement, EMG, Electromyography, Viewport Control

1 MOTIVATION

Head rotation is central to the interaction model of head-mounted displays (HMDs). HMDs provide a view into a 3D environment controlled by the head to support interaction as in the real world by orienting the head and torso per our tasks. Neck muscles drive head rotation and play an important part in HMD interaction. To achieve yaw or pitch head movements, neck muscles like the semispinalis capitis (at extensor muscle group), splenius capitis muscle (SPL), and sternocleidomastoid muscle (SCM) are deeply involved [16]. Besides, as a connection between the head and torso, the neck muscles play a crucial role in the detection of the orientation of the head [6, 12]. Together with visual and vestibular inputs, they are essential for the perception of the body in space [2, 7, 9, 6, 12]. Despite the significant involvement of neck muscles in HMD interaction, there are relatively few studies that explore how these muscles can influence such interactions. Thus, we raise research questions below:

- How do neck muscles affect HMD interaction?
- How to use neck muscle knowledge to improve HMD interaction?

Theoretically, insight into neck muscles in HMD has the potential to ease below typical HMD challenges.

Head amplification motion sickness VR motion sickness is mainly induced by sense conflict [11]. In particular, the sense conflict between visual and vestibular inputs (inner ear)[11]. Amplifying the display easily decouples vision and motion, but it is difficult to affect the vestibular system's sense of motion, making the issue difficult to address.

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However, neck muscles also participate in the human perception of the body in space [2, 7, 9, 6, 12]. In particular, the muscle spindle, a sensory receptor inside the muscle, primarily detects changes in the length of the muscle to feel motion [3, 13]. A simple demonstration of muscle spindle sense is closing your eyes and moving your hand in mid-air. Even without visual input, muscle spindles in your muscles provide feedback about changes in muscle length, allowing you to sense your hand's position roughly. This is also similar to the neck muscles that can sense our head position related to the torso. However, the muscle spindle is only sensitive to changes in muscle length and the speed of those changes instead of the absolute mapping between motion and muscle length [13]. In other words, the muscle spindle knows the motion is happening but doesn't precisely know how much it moves. For example, in neck muscles, when experiencing high neck strain, we expect to see further away, but we don't know this neck strain level should correlate to which head position. This character provides an opportunity to manipulate or decouple sensory motion and real motion.

Previous studies often used static gain to amplify head rotation, but the static gain amplification does not adapt to the change in neck muscle length. For instance, prior study shows static gain creates significantly more motion sickness compared to 1:1 mapping [10]. Specifically, if we amplify the head rotation according to the neck muscle length level, it probably makes visual input more convincing to the brain and then eases the motion sickness. This is because the brain often prioritizes visual information [14]. When the vision is strengthened by other proprioceptive integration like neck muscle sense, the brain may filter out unreliable sensory inputs like the vestibular system and tend to believe the vision [5]. In this way, the sense of neck muscle length and vision support each other, constructing positive feedback.

Neck muscle fatigue Interaction in HMDs induces a significant workload on the neck muscles to move and stabilize the head, including the additional weight of the HMD [4, 15]. HMDs have been shown to lead to increased head movements (due to the decreased field of view of HMDs) and higher load on the muscles (due to the additional weight of HMDs) [1]. This has been observed in head pointing [17] as well as prolonged usage (60 mins) with the content presented centrally in the field of view [8]. Understanding neck muscle strain in different head positions can help ease the issue, such as designing head amplification techniques to avoid moving the head to or maintaining high neck strain positions.

2 AIMS

This PhD research aims to investigate how neck muscle can affect HMD interaction and how to use this understanding to improve interaction experience. Current research mainly focuses on improving neck ergonomics and reducing head motion sickness caused by head amplification during HMD interactions.

3 METHODOLOGY

We plan to first deepen our understanding of the neck muscles during head movements. Considering muscle length is deeply interconnected with muscle strain feeling and muscle activity, we will use surface electromyography (sEMG) to quantify neck muscle strain in various head positions and for different holding durations.

Next, we will analyze and model the collected data. From this analysis, we will try to improve existing HMD interactions or solve some issues.

Finally, we will conduct user studies to evaluate the new interaction techniques' experience and determine whether they improve neck ergonomics compared to the baseline.

4 PROJECT PROGRESS

We have successfully leveraged the neck muscle understanding to improve head amplification. Specifically, we first conducted a study (N=15) to quantify neck muscle activity using sEMG. Users were tasked to shift their heads from neutral to indicated target positions and remain there for 30 seconds. Neck muscle activity was measured using surface electromyography (sEMG), with electrodes positioned on six areas on the side and back of the neck. Data was collected for 67 head poses over $\pm 80^\circ$ horizontally (yaw) and $\pm 50^\circ$ vertically (pitch). With this data, we trained an artificial neural network to build a model ($R^2 = 0.72$) that predicts neck muscle activity for any head fixation with $head_{yaw}, head_{pitch}, time$ as input. Finally, we designed the head amplification techniques and conducted a 360° video-watching task as an early test. The technique was evaluated in a study (N=18) compared with 1:1 mappings as a baseline and demonstrated to reduce head movement without compromising performance or increasing cybersickness. The above study is under CHI'25 review.

In the current stage, we can demonstrate that the proposed head amplification does not cause significant motion sickness when compared to 1:1 mappings. However, more studies are needed to further evaluate the effectiveness of this new head amplification technique. Additionally, further research is necessary to explore how other interaction techniques and methods can be refined and improved.

5 CONCLUSION

Previous studies point out that remapping motion and vision (head amplification) in VR reduces physical workload but increases mental workload (motion sickness induced) [11]. The use of static gain amplification highlights a clear conflict between achieving higher head amplification and reducing motion sickness, placing head amplification in a challenging dilemma [10]. By using insight into the neck muscles, we have the potential to ease the conflict between physical and mental ergonomics.

Besides, exploring the neck muscle effect in HMD interaction provides a new perspective on improving user experience.

6 CURRENT CHALLENGES

The project is still unfinished but has a lot of potential for improvement. I have several areas where I would appreciate feedback.

- Can any other interaction be improved by neck muscle understanding? Are there any tricky issues that can be improved?
- Does my research on neck muscle make a meaningful contribution to VR or HCI? Does it create sufficient novelty?
- On the proposed head amplification, any follow-up/future study is necessary or interesting?

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